Lead 3: index calculus on cubic-extension curves (Weil descent, n = 3)

Branch ic-lead3-weil-n3-20261030. Instrument: repo `gaudry_cubic` (unmodified) + `fit_n3.rs`; counted unit = affine additions + Fp muls/63; rho pooled S = 1.336 ± 0.104 (30 instances). Measured values; derived values are fits (OLS, bootstrap 95%).

Literature. Gaudry–Diem decomposes R into n factor-base points (x in Fq), Weil-restricted Semaev S(n+1). With large primes the cost is O (q with exponent 2-2/n) (Joux–Vitse 2013 §2.1; Diem, Math. Comp. 80 (2011), Thm 3 per a search snippet; Gaudry JSC 2009 not read first-hand). At n = 3: q with exponent 4/3 (n with exponent 4/9) against rho at n with exponent 1/2. Joux–Vitse m = n-1: q squared, worse than rho for n ≤ 4.

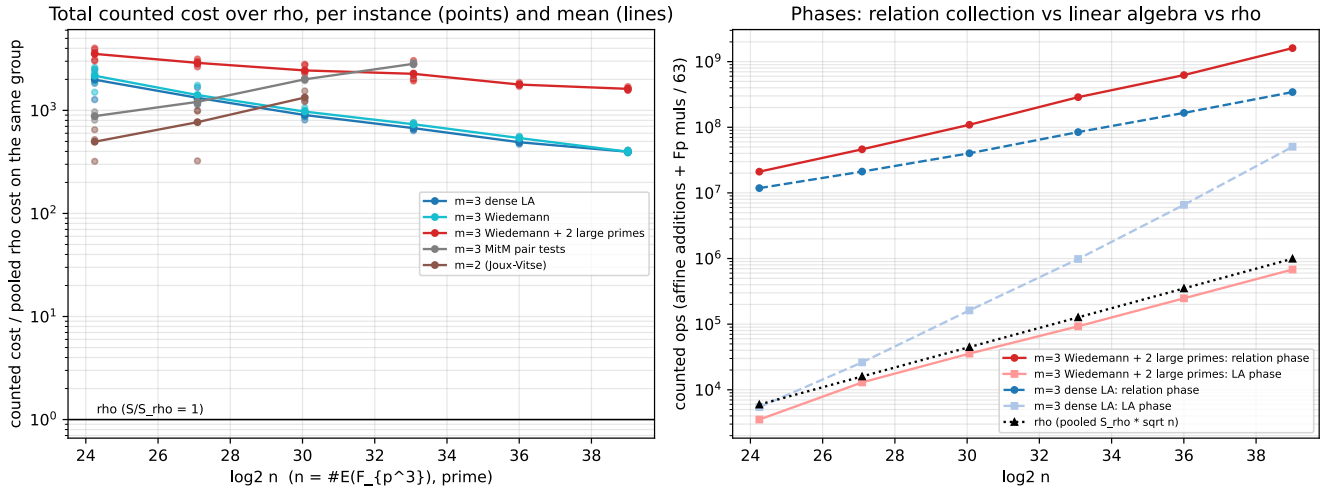


Figure 1: Left: counted cost / pooled rho, per instance (points) and mean (lines). Right: phases vs rho. p = 271..8209 (log2 n 24..39), 5 seeds (arm P), 3 seeds (MitM, m=2 to p = 2083; m=2 at p = 2083 censored at the 1e6-residual cap, excluded). Source: results.json.

arm	p	log2 n	residuals	ops/residual	cost / rho
2LP	271	24.2	1502	1.40e4	3535
2LP	8209	39.0	107863	1.49e4	1618
dense LA	8209	39.0	23077	1.49e4	397
Wiedemann	8209	39.0	25095	1.49e4	398
MitM m=3	2083	33.1	5591	6.4e4	≈2.9e3
m=2 (JV)	1039	30.1	523580	1.1e2	≈1.3e3

Fitted exponents in n (p ≤ 8209): 2LP total 0.425 [0.416, 0.434], relation phase 0.425, LA phase 0.505 [0.482, 0.525]; full-base relation 0.333, dense LA 0.894; MitM 0.697; theory 4/9, 1/3, 2/3.

Projection (derived, 140–190 bit extrapolation, indicative): 2LP two-phase model bottoms at ≈1.5× rho near 2^226 and never goes below 1; theory exponents give a crossover near 2^230; full-base arms bottom at ≈200–340× rho near 2^42–2^50. Pricing unpriced hash probes at 0/0.1/1 GAE does not change the order.

Verdict. The n = 3 exponent beats rho only for m = 3 with double large primes, and the measured cost is 1.6e3× rho at 2^39, falling slowly; no reachable crossover. Replicates and extends RESEARCH_RESIDUAL_WALKS §11.15; no new claim.